K is a break point for H if no dataset of size K can be Shattered by H

$$m_H(K) < 2^K$$

For 2D perceptron K = 4

if H can shatter atmost d points K = d + 1 is the break point.

In Hoeffding inequality

$$E_{out} \le E_{in} + \sqrt{\frac{1}{2N} \ln\frac{2M}{\delta}}$$

M - no of hypothesis

M is replaced with $m_H(N)$, which counts effective hypothesis (distinct behaviours on data), not all hypothesis.

$m_H(N)$ only helps if it is POLYNOMIAL

if $m_H(N) = 2^N$ (no break point)

$$\sqrt{\frac{1}{2N} \frac{\ln 2 \cdot (2^N)}{\delta}} \approx \sqrt{\frac{N}{N} \frac{\ln 2}{\delta}}$$

$\approx$ constant

if $m_H(N) \le N^{K-1}$ (polynomial)

$$\sqrt{\frac{\ln(2N^{K-1})}{2N \quad \delta}} = \sqrt{\frac{(K-1)\ln N}{2N}} \to 0$$

as $N \to \infty$

logarithm grows far slower than N

$$\boxed{\text{Sauers Theorem : If a break point K exists at all, then } m_H(N) \le \sum_{i=0}^{K-1} {}^N C_i \le N^{K-1} + 1}$$

B(N, K) is the maximum numbers of dichotomies on N points with break point K

$B(N, 1) = 1$

$B(1, K) = 2$

Suppose for x_1 x_2 x_3 ... x_{N-1} $(N \ge 2, K \ge 2)$ we have (+1, -1 ... +1) labeling and when we add point x_N we can get

Case A : S_1

$(+1, -1, \cdots +1, | +1)$ (x_N)

but H cannot produce

$(+1, -1, \cdots +1, | -1)$

So In this case N-1 point has only one extension

Case B : S_2

The H produces

$(+1, -1, \cdots +1, | +1)$

and $(+1, -1 \cdots +1, | -1)$

	no of Rows	x_1 $x_2 \cdots x_{N-1}$	x_N
S_1	α	+1 +1 ··· +1 -1 +1 ·· +1 ⋮	+1 -1 ⋮ -1 +1
S_2^+	β	+1 -1 ·· +1 -1 -1 ··· +1 -1 -1 ··· -1	+1 +1 ⋮ +1
S_2^-	β	+1 -1 ·· +1 -1 -1 ·· +1 ⋮ -1 -1 ·· -1	-1 -1 ⋮ -1

α = Number of (N-1) point labelings that only have one possible label for x_N

β = No of (N-1) point labelings with +1 and -1 for x_N

$$B(N, K) = \alpha + 2\beta$$

For N-1 points, the total different dichotomies is given by $\alpha + \beta$ as for N-1 points S_2^+ and S_2^- are the same dichotomies

$$\alpha + \beta \le B(N-1, K)$$

$S_2 = S_2^+ \cup S_2^-$, S_2 contains patterns that can be extended with both $x_N = +1$ and $x_N = -1$, suppose those β patterns on N-1 points could realise every possible labeling of some K-1 points. Then because each one can be extended in both (+, -) ways at x_N, we could use x_N and realize every labeling of K points

But K is break point

$$B(K, K) < 2^K$$

So S_2 must have break point of K-1

$$\beta \le B(N-1, K-1)$$

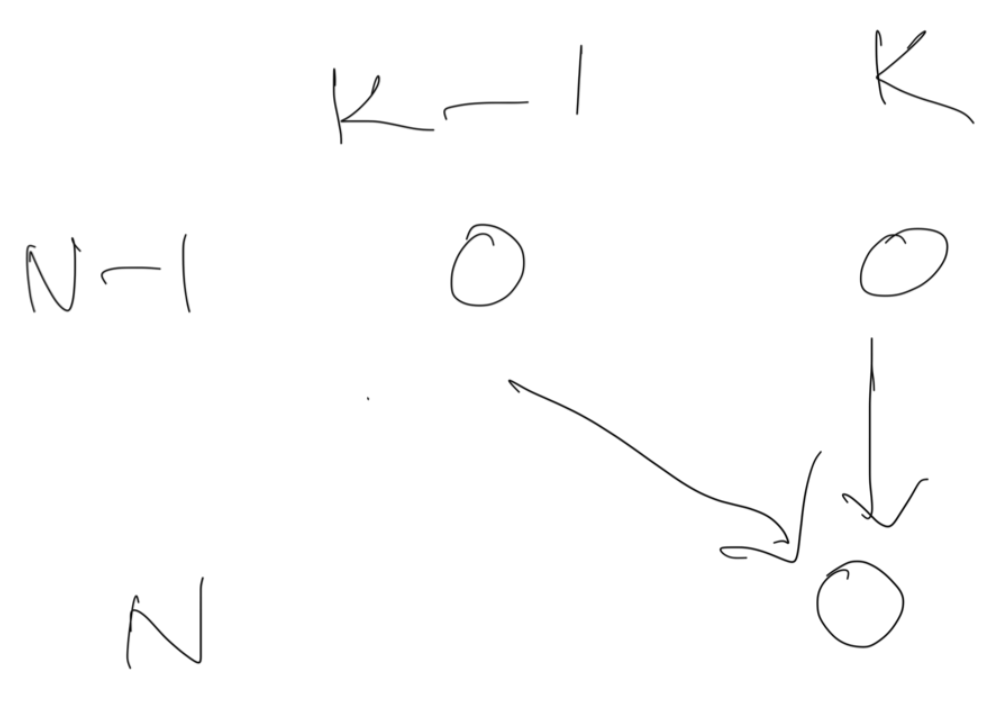
$$B(N, K) = \alpha + 2\beta$$
$$= \alpha + \beta + \beta$$

$$\alpha + \beta \le B(N-1, K)$$
$$\beta \le B(N-1, K-1)$$

$$\Rightarrow \boxed{B(N, K) \le B(N-1, K) + B(N-1, K-1)}$$

$${}^N C_i = {}^{N-1} C_i + {}^{N-1} C_{i-1}$$

			K				
		1	2	3	4	5	6 ···
N	1	1	2	2	2	2	·
	2	1	3	4	4	4	·
	3	1	4	7	8	8	· ·
	4	1	5	11	15	16	· ·
	5	1	6	16	26	···	
	6	1	7	22	···		
	⋮	⋮					



$$B(N_0+1, K) \le B(N_0, K) + B(N_0, K-1)$$

$$\boxed{\text{Sauer's Lemma} \quad B(N, K) \le \sum_{i=0}^{K-1} ({}^N C_i)}$$

$$B(N_0, K) \le \sum_{i=0}^{K-1} ({}^{N_0} C_i)$$

$$B(N_0, K-1) \le \sum_{i=0}^{K-2} ({}^{N_0} C_i)$$

$$B(N_0+1, K) \le \sum_{i=0}^{K-1} ({}^{N_0} C_i) + \sum_{i=0}^{K-2} ({}^{N_0} C_i)$$

$$\sum_{i=0}^{K-1} {}^{N_0} C_i = {}^{N_0} C_0 + {}^{N_0} C_1 + \cdots {}^{N_0} C_{K-1}$$
$$= 1 + {}^{N_0} C_1 + \cdots {}^{N_0} C_{K-1} = 1 + \sum_{i=1}^{K-1} ({}^{N_0} C_i)$$

$$\sum_{i=0}^{K-2} {}^{N_0} C_i = {}^{N_0} C_0 + {}^{N_0} C_1 + \cdots {}^{N_0} C_{K-2}$$
$$= \sum_{i=1}^{K-1} {}^{N_0} C_{i-1}$$

$$\Rightarrow B(N_0+1, K) \le 1 + \sum_{i=1}^{K-1} ({}^{N_0} C_i) + \sum_{i=1}^{K-1} ({}^{N_0} C_{i-1})$$

$$1 + \sum_{i=1}^{K-1} \left[({}^{N_0} C_i) + ({}^{N_0} C_{i-1})\right]$$

$$1 + \sum_{i=1}^{K-1} ({}^{N_0+1} C_i)$$

As ${}^{N_0+1} C_0 = 1$

$$\sum_{i=0}^{K-1} ({}^{N_0+1} C_i)$$

$$B(N_0+1, K) \le \sum_{i=0}^{K-1} ({}^{N_0+1} C_i)$$

Example For K=3

$$B(N, 3) \le {}^N C_0 + {}^N C_1 + {}^N C_2$$

if N=5

$$B(5, 3) \le 1 + 5 + 10$$
$$\le 16$$

The max is $2^N = 2^5 = 32$

A class with breakpoint K=3 can produce a maximum of 16 dichotomies not 32

$$B(N, K) \le \sum_{i=0}^{K-1} ({}^N C_i)$$

$${}^N C_i = \frac{N(N-1)(N-2) \cdots (N-i+1)}{i!}$$

$$= O(N^i)$$

$B(N, K) = \sum {}^N C_i = O(N^{K-1})$ K-1 = largest degree = polynomial

without break point

$m_H(N) = 2^N$ = exponential

with break point K, we get a upper bound

$$m_H(N) \le B(N, K) \le \sum_{i=0}^{K-1} {}^N C_i$$

For a given H, K is fixed. we get a polynomial bound on $m_H(N)$

Examples : positive rays

K = 2

$$m_H(N) \le {}^N C_0 + {}^N C_1 \le N + 1$$

positive Interval

K = 3

$$m_H(N) \le {}^N C_0 + {}^N C_1 + {}^N C_2 = 1 + N + \frac{N^2}{2}$$

2D perceptron

K = 4

$$m_H(N) \le {}^N C_0 + {}^N C_1 + {}^N C_2 + {}^N C_3$$
$$\le \frac{N^3}{6} + \frac{5N}{6} + 1$$